

Information and Communication Technology (CS407)
M.C.A 1st Semester
Autumn Semester End-Term Examination 2017
Computer Science and Engineering

TU/CSE

Time Duration: 2 Hours

Total Marks:60

Note: Please write your answers precisely and to the point.

1. Choose the correct option

1X15 = 15 Marks

- (a) The method of communication in which transmission takes place in both directions, but only in one direction at a time is called _____
(simplex / half simplex / half duplex / full duplex)
- (b) End-to-end connectivity with reliable delivery of message is provided by the _____ layer. (network / transport / session / data-link)
- (c) Which of the layer in the OSI model is more closely related to the physical communication facilities ? (Application / Session / Network / Data-link)
- (d) In a multi-programming environment _____
(the processor executes more than one process at a time / the programs are developed by more than one person / more than one process resides in the memory / a single user can execute many programs at the same time.)
- (e) In paged memory systems, if the page size is increased, then the internal fragmentation generally _____
(becomes less / becomes more / remains constant / none of the options)
- (f) Fragmentation is _____
(dividing the secondary memory into equal sized fragments / dividing the main memory into equal-sized fragments / fragments of memory used in a page / fragments of memory words unused in a page)
- (g) Virtual memory is _____
(an extremely large main memory / an extremely large secondary memory / an illusion of an extremely large memory / a type of memory used in super computer computers)
- (h) Page fault occurs when _____
(the page is corrupted by application software / the page is in main memory / the page is not in main memory / one tries to divide a number by 0)
- (i) A process, if in a 'BLOCKED' state waiting for some I/O service, the process goes to the _____ state when the service is completed.
(RUNNING / READY / SUSPENDED / TERMINATED)
- (j) Thrashing _____
(reduces page I/O / decreases the degree of multiprogramming / implies excessive page I/O / improves the system performance)
- (k) The memory interleaving technique is mainly used to reduce the speed

mismatch between _____.

(processor and cache memory / processor and main memory / cache memory and virtual memory / cache memory and main memory)

- (l) It is best to opt for _____ to set up communication line within a local or small area.
(co-axial cable / UTP / both co-axial and UTP / none of the options)
- (m) There are _____ incompatibility issues when forming an Internet out of networks. (both hardware and software / only hardware / only software / software and firmware)
- (n) When except other party, only both sender and receiver can access the contents of a message, the _____ principles comes into picture.
(confidentiality / authentication / authorization / integrity)
- (o) Digital signature is done by using _____.
(only private key of the sender / only public key of the sender / private key of the receiver / none of the options)

2. State True or False

1X15 =15 Marks

- (a) A router is a special-purpose computer that is used specifically for → True
networking purposes.
- (b) Just one address is sufficient for a router. → True
- (c) A repeater is more intelligent than a bridge. → True
- (d) An IP address of a computer is also known as logical address mainly because of its attachment to a particular or multiple network in Internet.
- (e) Address resolution is a process used for obtaining the physical address of a → True
computer based on its IP address.
- (f) The objective of Digital Signature is mainly to disallow non-repudiation. → True
- (g) Bridge in a network is recognized using IP address.
- (h) Bridges know the physical addresses of different hosts on the network.
- (i) Paging is just another method to solve external fragmentation problem by making scattered free spaces contiguous.
- (j) Loosely coupled multiprocessing system means all the processors are not → True
bound each other and have their own local memory.
- (k) The smallest unit in any mechanical disk is track. → True
- (l) A cylinder in a disk is formed by the adjoin of subsequent tracks in each surface of each disk.
- (m) A socket is a combination of 48 bits MAC address and port number.
- (n) The hashing algorithm used to generate message digests is mainly to validate → True
the authenticity of the original sender's message.
- (o) The masking bits from within the host bits gives idea of only the number of → True
subnets to be created.

3. Write pointwise how SSL using Digital Certificate guarantees a secure communication between two parties.

5 Marks

5. Write the steps of how Digital Signature guarantee the authenticity of a sender when sending a document to a receiver. 7 Marks

6. For the IP address 160.139.219.242/4 of a class based network. Find the following. $1+2+1\times 2+2\times 2+2\times 2=13$ Marks

- (a) Class of the IP address
 - (b) Subnet Mask
 - (c) Total number of subnet and hosts per subnet
 - (d) Network Address and Broadcast address of the current IP address 160.139.219.242
 - (e) Network Address and Broadcast address of the 5th subnet
6. Assume you 've a main memory of 100 bytes and the pages of a process of 24 bytes is occupying the memory currently in a distributed fashion. If the size of your page is 4 bytes and the CPU issues a logical address of Page Number as 5 and Page Offset as 2 which is in Frame Number 12. Find out the following.

- (a) The number of Pages that 'll be required by the Process.
 - (b) The actual Physical address in the main memory to access. Assume that memory location is starting from location 0000000 and Frame number starting from Frame 0.
- $2+3 = 5$ Marks